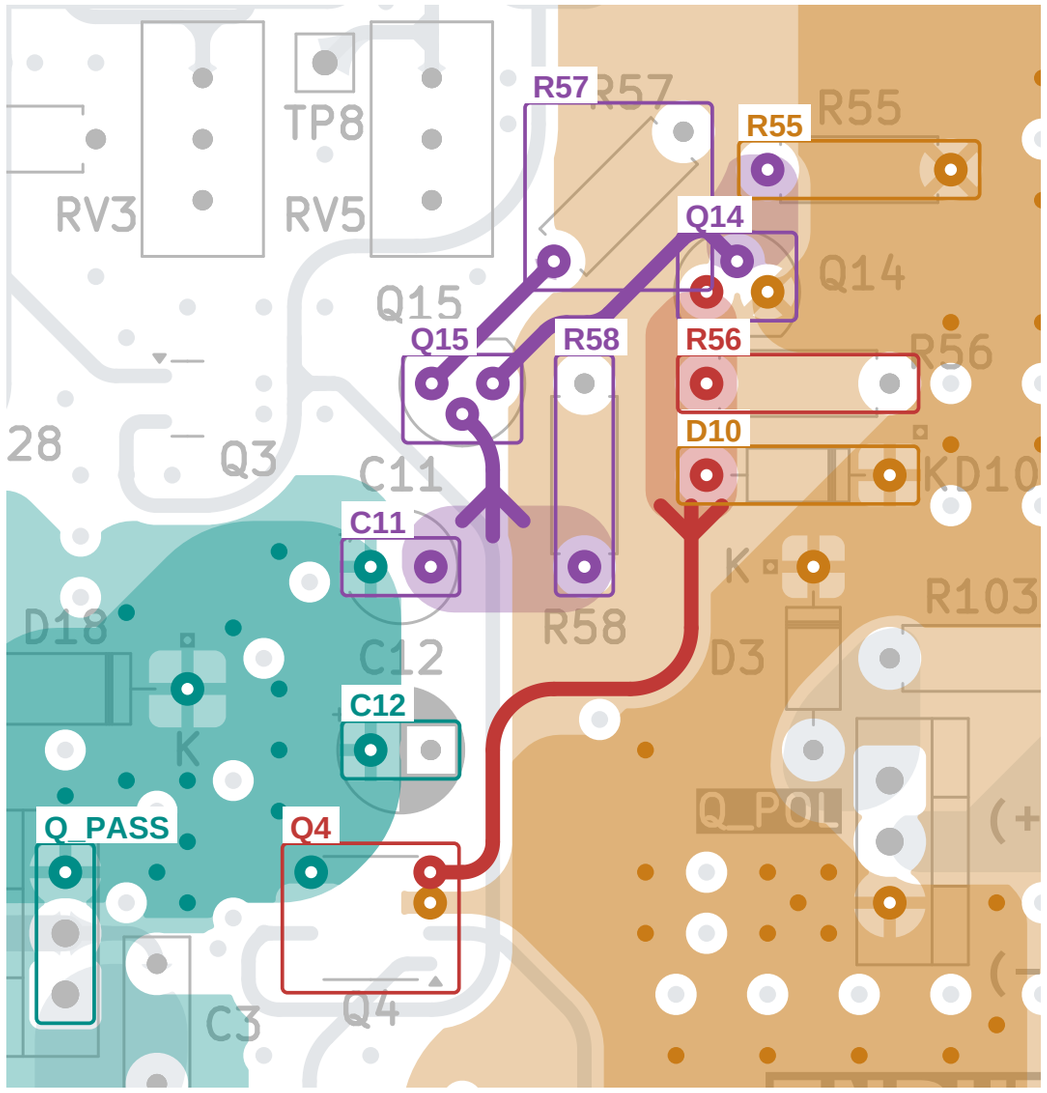
Startup ramp and series switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers



Protected supply
VEE supplies the startup stage and Q4B pin 3.

Ramp feedback
C11 couples changes of VCTRL into the Q15 / Q14 control network.

Switch gate
The Q14-controlled node drives Q4B gate pin 4; D10 connects it to VEE.

Ramped supply
Q4B pin 5 feeds VCTRL, C12 and the pass emitter. C12 is 10 uF in the PCB.

Q4B is the startup section of the shared Q4 package. The enable section Q4A is a separate path.

VEE → Q4B pins 3-5 → VCTRL / C12 → C11 / Q15 → Q14 → Q4B gate

The first three blocks show the supply path; the remaining blocks show the feedback path to the switch gate.

Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.